

HERVEY BAY, Australia

WMO: 955650

Lat: 25.322S

Lon: 152.882E

Elev: 43

StdP: 14.67

Time Zone: 10.00 (E10)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.6	45.4	30.7	24.9	60.6	34.4	29.2	57.4	18.5	65.8	16.7	66.1	3.1	210	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	87.6	74.2	86.3	73.8	85.1	73.3	78.7	83.3	77.4	82.2	76.4	81.5	13.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.3	81.5	76.0	136.2	80.7	74.9	130.8	79.9	42.3	83.5	41.0	82.5	40.0	81.4	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	18.9	17.0	DB	37.1	90.9	2.6	1.5	35.2	92.0	33.7	92.8	32.2	93.7	30.3	94.8	(4)
(5)			WB	35.8	82.1	2.5	3.3	34.0	84.5	32.6	86.4	31.2	88.3	29.5	90.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	78.7	78.4	76.1	71.8	65.9	61.8	60.1	61.3	66.1	70.4	74.1	77.1	(6)	
	DBStd	7.45	2.78	2.61	2.56	2.67	3.73	4.07	3.92	3.91	4.01	3.69	3.37	3.02	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	463	0	0	0	1	35	108	156	128	32	3	0	0	(9)	
	CDD50	7337	890	796	809	653	491	356	314	349	484	632	724	839	(10)	
	CDD65	2326	425	376	344	204	61	14	5	12	66	170	275	374	(11)	
	CDH74	16161	3805	3245	2395	914	175	10	5	24	204	722	1666	2996	(12)	
	CDH80	3160	982	835	435	57	2	0	0	1	5	31	189	623	(13)	
(14) Wind	WSAvg	8.7	10.6	10.1	9.6	8.6	7.3	7.0	6.5	7.0	8.0	9.3	9.9	10.2	(14)	
(15) Precipitation	PrecAvg	46.3	6.2	7.6	6.1	3.5	3.7	2.9	2.0	1.9	1.8	2.7	3.1	4.8	(15)	
	PrecMax	67.5	16.8	18.7	18.3	6.8	10.8	8.6	7.2	9.0	5.1	8.2	5.6	21.3	(16)	
	PrecMin	31.5	0.7	2.5	1.7	0.9	0.0	0.8	0.5	0.0	0.1	0.4	0.7	1.1	(17)	
	PrecStd	9.8	3.5	4.4	3.8	1.4	2.6	2.0	1.5	2.1	1.5	1.8	1.3	3.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.1	89.2	87.5	83.3	80.0	75.3	74.6	76.2	80.2	82.8	85.6	88.7	(19)	
		MCWB	75.3	74.7	74.6	71.6	67.8	65.6	62.6	63.0	66.2	67.3	70.3	70.9	(20)	
	2%	DB	87.4	87.6	85.5	81.6	77.8	73.5	72.8	74.3	77.5	80.6	83.5	85.9	(21)	
		MCWB	74.3	74.8	73.7	70.5	66.4	64.3	61.6	63.0	65.5	69.0	70.8	73.8	(22)	
	5%	DB	86.1	86.2	84.0	80.3	76.0	71.9	71.3	72.8	76.3	79.2	82.1	84.6	(23)	
		MCWB	73.8	74.2	72.9	69.5	65.9	63.1	61.1	62.0	65.2	68.4	70.0	73.0	(24)	
	10%	DB	84.9	84.8	82.6	79.0	74.1	70.2	69.5	71.4	74.9	77.9	80.8	83.5	(25)	
		MCWB	73.3	73.6	72.1	69.1	65.1	62.5	60.6	61.1	64.7	67.7	69.5	72.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.3	80.1	79.0	78.0	72.5	69.1	68.4	68.7	71.9	75.2	76.3	79.8	(27)	
		MCDB	85.9	84.9	82.8	79.9	76.3	71.8	70.8	72.5	75.8	77.1	80.7	83.4	(28)	
	2%	WB	78.4	78.2	76.8	74.0	69.8	67.5	66.4	67.3	70.0	73.2	74.6	77.9	(29)	
		MCDB	83.1	83.7	81.0	77.9	74.1	70.6	69.7	71.2	74.8	77.3	80.0	82.4	(30)	
	5%	WB	77.2	77.0	75.5	72.0	68.1	65.8	64.4	65.7	68.7	71.7	73.3	76.3	(31)	
		MCDB	82.3	82.5	80.6	76.9	73.5	69.6	68.1	69.8	73.4	76.3	78.9	81.3	(32)	
	10%	WB	76.1	75.9	74.4	70.7	66.6	64.3	62.5	63.8	67.6	70.2	72.0	74.9	(33)	
		MCDB	81.6	81.6	79.9	76.3	71.9	68.7	67.5	68.7	72.3	75.3	78.1	80.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.0	12.5	14.2	16.6	16.7	19.4	20.4	17.6	14.6	12.9	11.8	(35)	
		MCDBR	12.8	13.9	14.6	15.9	18.0	17.3	20.6	21.0	18.4	15.7	13.6	13.2	(36)	
	5% WB	MCWBR	4.2	4.7	5.5	6.7	8.8	9.4	11.2	11.7	9.3	6.7	5.2	5.2	(37)	
		MCWBR	9.9	11.0	11.2	13.0	15.1	13.9	16.0	15.7	13.4	10.9	10.4	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.389	0.384	0.369	0.344	0.324	0.318	0.308	0.319	0.353	0.381	0.376	0.391	(40)	
		taud	2.530	2.550	2.586	2.641	2.644	2.657	2.664	2.619	2.513	2.464	2.516	2.502	(41)	
	Ebn at Noon		303	300	295	289	282	278	285	293	295	297	305	303	(42)	
		Edh at Noon	36	34	31	27	25	23	24	27	33	37	36	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2116	1920	1680	1467	1224	1047	1191	1475	1806	1983	2152	2151	(44)	
	RadStd		155	168	131	107	61	80	101	109	129	156	190	225	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (15 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-144	+374	+236

Nomenclature: See separate page